SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME:CSA07

COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Y Siddartha/192e425047 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

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Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of

retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoTenabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, wellexplained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

Organization	15%	Wellstructured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answer

1. Transport and Data Link Layer Calculation

Given

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: Convert file size to bytes

150 MB = 150×10^6 = **150,000,000 bytes**

Step 2: Number of segments

Number of segments = 150,000,000 / 1500 = **100,000 segments**

Step 3: Number of frames

Each segment becomes one frame.

Frames transmitted = 100,000

Step 4: Total Data Link overhead

Overhead = 100,000 × 40 = **4,000,000 bytes = 4 MB**

Step 5: Total data transmitted

Total = 150,000,000 + 4,000,000 = **154,000,000 bytes**

Convert to bits:

154,000,000 × 8 = **1,232,000,000 bits**

Step 6: Transmission time

Time = 1,232,000,000 / 50,000,000 = **24.64 s**

Answer

- Segments = **100,000**
- Frames = **100,000**
- Data Link overhead = **4 MB**
- Transmission time ≈ **24.64 s**

Reliability

- **Transport Layer:** sequencing, acknowledgments, retransmission, flow control.
- **Data Link Layer:** framing, MAC addressing, error detection using CRC, and retransmission of corrupted frames.

2. Sliding Window Protocol

Given

- Window size = **6**
- Total frames = **48**
- Frame size = **1024 bytes**

Step 1: Number of windows

Windows = 48 / 6 = **8 windows**

Step 2: Retransmissions

One frame is lost in every window.

Retransmissions = **8**

Answer

- Transmission windows = **8**
- Total retransmissions = **8 frames**

Flow Control

Flow control prevents the sender from overwhelming the receiver, improves buffer utilization, reduces packet loss, and increases overall throughput.

3. IP Fragmentation

Given

- IP packet = **12,000 bytes**
- MTU = **1500 bytes**
- IP header = **20 bytes**

Step 1: Payload per fragment

Payload = 1500 − 20 = **1480 bytes**

Step 2: Number of fragments

12,000 / 1480 = 8 remainder 160

Fragments = **9**

Step 3: Header overhead

Overhead = 9 × 20 = **180 bytes**

Answer

- Fragments created = **9**
- Header overhead = **180 bytes**

Network Layer Function

The Network Layer assigns fragment offsets and identification fields so that the destination host can correctly **reassemble all fragments** into the original packet.

4. Sliding Window (same as Question 2)

Answer

- Windows required = **8**
- Retransmissions = **8 frames**

Flow Control Benefit

It matches the sender's transmission rate with the receiver's processing capacity, reducing congestion and improving communication efficiency.

5. Session Layer Synchronization

Given

- Video size = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Step 1: Checkpoints before failure

390 / 60 = **6 full checkpoints**

Checkpoints at: 60, 120, 180, 240, 300, 360 MB

Step 2: Data to retransmit

Failure occurs at 390 MB.

Last checkpoint = 360 MB

Retransmission = 390 − 360 = **30 MB**

Answer

- Checkpoints created = **6**
- Data retransmitted = **30 MB**

Importance

Synchronization allows communication to **resume from the last checkpoint instead of restarting the entire transfer**, saving bandwidth and time.

6. Presentation Layer Compression and Encryption

- **Given**
- Original file = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%**

Step 1: Compressed size

Compressed = 900 × (1 − 0.40) = **540 MB**

Step 2: Encrypted size

Encrypted = 540 × 1.05 = **567 MB**

Step 3: Percentage reduction

Reduction = 900 − 567 = **333 MB**

Percentage reduction = (333 / 900) × 100 = **37%**

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Presentation Layer Role

- **Compression:** reduces transmission size and bandwidth usage.
- **Encryption:** provides confidentiality and protects sensitive data during transmission.

7. FTP Application Transfer

Given

- Data = **3 GB**
- Throughput = **120 Mbps**
- Request size = **3 MB**

Step 1: Convert data

3 GB = 3000 MB

Step 2: Number of requests

Requests = 3000 / 3 = **1000 requests**

Step 3: Convert data to bits

3000 MB × 8 = **24,000 Mb**

Step 4: Transmission time

Time = 24,000 / 120 = **200 s**

Answer

- Requests generated = **1000**
- Transmission time ≈ **200 s**

Application Layer Services

The Application Layer provides file transfer services, user authentication, error reporting, session management, and reliable interaction between users and network applications.

8. End-to-End Delay

Given

- Application = 4 ms

- Transport = 3 ms
- Network = 5 ms
- Data Link = 2 ms
- Physical = 1 ms
- Propagation = 18 ms
- Transmission = 12 ms

Step 1: Processing delay

Processing = 4 + 3 + 5 + 2 + 1 = **15 ms**

Step 2: Total delay

Total = 15 + 18 + 12 = **45 ms**

Answer

Total end-to-end delay = 45 ms

Layer Contributions

- **Application:** prepares user data.
- **Transport:** segmentation and reliability.
- **Network:** routing decisions.
- **Data Link:** framing and error detection.
- **Physical:** converts bits to signals.
- **Transmission delay:** time to place bits on the medium.
- **Propagation delay:** time for signals to travel through the network.

9. Protocol Overhead and Efficiency

Given

- Useful data = **80 MB**
- Frame size = **1600 bytes**
- Overhead per frame = **80 bytes**

Step 1: Useful payload per frame

Payload = 1600 − 80 = **1520 bytes**

Step 2: Convert useful data

80 MB = 80×10^6 = **80,000,000 bytes**

Step 3: Number of frames

Frames = 80,000,000 / 1520 ≈ **52,632 frames**

Step 4: Total overhead

Overhead = 52,632 × 80 = **4,210,560 bytes ≈ 4.21 MB**

Step 5: Transmission efficiency

Efficiency = (1520 / 1600) × 100 = **95%**

Answer

- Frames required ≈ **52,632**
- Total overhead ≈ **4.21 MB**
- Transmission efficiency ≈ **95%**

Effect of Protocol Overhead

Higher overhead consumes additional bandwidth, increases transmission time, reduces effective throughput, and lowers overall network efficiency.

10. Multi-Layer OSI Communication

Given

- Medical record = **240 MB**
- Compression = **30%**

- Segment size = **1400 bytes**
- Data Link overhead = **60 bytes/frame**

Step 1: Compressed file size

Compressed = 240 × 0.70 = **168 MB**

168 MB = **168,000,000 bytes**

Step 2: Number of segments

Segments = 168,000,000 / 1400 = **120,000 segments**

Step 3: Protocol overhead

Overhead = 120,000 × 60 = **7,200,000 bytes = 7.2 MB**

Step 4: Final data transmitted

Final = 168,000,000 + 7,200,000 = **175,200,000 bytes**

= **175.2 MB**

Answer

- Compressed file size = **168 MB**
- Segments transmitted = **120,000**
- Data Link overhead = **7.2 MB**
- Final transmitted data = **175.2 MB**

How OSI Layers Work Together

- **Presentation Layer:** compresses and encrypts medical data.
- **Transport Layer:** divides data into ordered segments and ensures reliable delivery.

- **Network Layer:** routes packets across the IoT healthcare network.
- **Data Link Layer:** frames the segments and performs error detection.
- **Physical Layer:** transmits the final bit stream through the communication medium.

This cooperation ensures **secure, efficient, and reliable transfer of sensitive patient records**.